Adding & Subtracting Fractions

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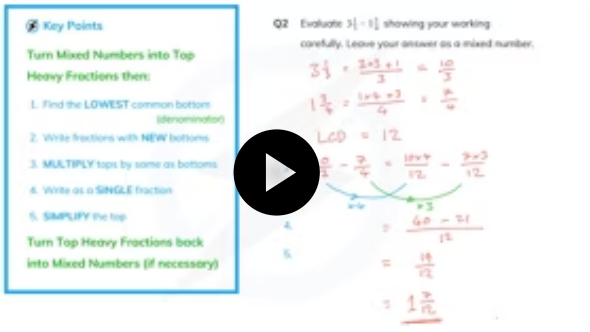

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Adding & subtracting fractions

How do I add or subtract two fractions?

- **Addition** and **subtraction** are treated in exactly the same way when dealing with fractions
  - Find the **lowest common denominator**
  - Write each fraction as an **equivalent fraction** over this denominator
  - **Add** (or **subtract**) the **numerators** and write this over a **single** denominator
    - Do **not** add the denominators
  - **Simplify** the fraction by cancelling common factors

What do I do if any of the fractions are mixed numbers?

- **Convert any** mixed numbers into **improper fractions**
  - Add or subtract the fractions as usual
  - Remember to **convert the answer** back to a mixed number if required

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Worked Example

(a) Find $\frac{2}{3} + \frac{1}{5}$.

Answer:

Find the lowest common denominator of 3 and 5
15 is the smallest number that can be divided by both 3 and 5

The lowest common denominator is 15

Write both fractions as equivalent fractions with a common denominator of 15
Multiply the numerator and denominator of the first fraction by 5
Multiply the numerator and denominator of the second fraction by 3

$$\frac{2 \times 5}{3 \times 5} + \frac{1 \times 3}{5 \times 3} = \frac{10}{15} + \frac{3}{15}$$

Add the numerators and write over a single denominator

$$\frac{10 + 3}{15} = \frac{13}{15}$$

There are no common factors that will cancel

$$\frac{13}{15}$$

(b) Find $3\frac{3}{4} - \frac{5}{8}$ giving your answer as a mixed number.

Answer:

Change the mixed number into an improper fraction

Multiply the denominator by the whole part and add to the numerator
Write this value over the denominator

$$\frac{4 \times 3 + 3}{4} = \frac{15}{4}$$

Find the lowest common denominator of 4 and 8
8 is the smallest number that by divided by both 4 and 8
The lowest common denominator is 8

Write both fractions as equivalent fractions over 8
Multiply the numerator and denominator of the first fraction by 2

$$\frac{15 \times 2}{4 \times 2} - \frac{5}{8} = \frac{30}{8} - \frac{5}{8}$$

Subtract the numerators and write over a single denominator

$$\frac{30 - 5}{8} = \frac{25}{8}$$

Change into a mixed number
Divide 25 by 8 to get 3 remainder 1

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